

SMART COMMUNITY SYSTEM

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1. Introduction

Smart Community is a mobile-based campus incident reporting system designed to help students and administrators improve campus safety. The system allows users to report incidents, upload images, receive notifications, and track the status of reports in real time. The system also includes an admin dashboard for managing reports, users, and notifications.

2. System Requirements

Hardware Requirements

- Android smartphone or emulator
- Stable internet connection

Software Requirements

- Firebase Authentication
- Firebase Realtime Database
- Cloudinary Image Storage
- Android Studio (for developers)

3. User Roles

The system has two main roles:

Student/User

Users can:

- Register account
- Login account
- Submit incident reports
- Upload images
- View notifications
- Track reports
- Edit profile
- Switch dark/light mode

Administrator

Admins can:

- View all reports

- Manage reports
- Update report status
- View users
- View user logs

4. User Registration

Steps to Register

1. Open the SmartCommunity application.
2. Tap the “Register” button.
3. Enter the following information:
 - Full Name
 - Email Address
 - Password
 - Confirm Password
4. Tap “Create Account”.

Registration Process

- The system validates the entered information.
- Firebase Authentication creates the user account.
- User information is automatically saved to Firebase Realtime Database.
- The user role is automatically assigned as “Student”.

Successful Registration

After successful registration, the user will be redirected to the login page.

5. User Login

Steps to Login

1. Open the application.
2. Enter:
 - Email
 - Password
3. Tap “Login”.

Login Process

- The system verifies credentials using Firebase Authentication.
- User activity logs are automatically recorded.
- Users are redirected based on their role.

User Redirect

- Student → Homepage Dashboard
- Admin → Admin Dashboard

6. Homepage Dashboard

The homepage contains the following modules:

-Report Incident

Allows users to submit incidents.

-Reports

Displays all submitted reports.

-Notifications

Displays system notifications and report updates.

-Profile

Displays user information.

- Logout

Logs the user out from the system.

7. Report Incident Module

Purpose

The Report Incident module allows users to report incidents happening inside the campus.

Report Information

Users must provide:

- Incident Title
- Description
- Location
- Category

- Image Evidence

Categories

Examples:

- Theft
- Accident
- Suspicious Activity
- Emergency

Report Submission Process

1. Fill all required fields.
2. Upload or capture an image.
3. Tap “Submit Report”.

After Submission

- Report is saved to Firebase Database.
- Image is uploaded to Cloudinary.
- Notification is automatically generated.
- User logs are automatically recorded.

8. Uploading Images

The system supports:

- Gallery image selection
- Camera image capture

Steps

1. Tap “Upload Image”.
2. Choose:
 - Camera
 - Gallery
3. Select or capture image.
4. The image preview will appear.
5. Image is uploaded to Cloudinary during report submission.

9. Smart Incident Priority

The Smart Priority System automatically determines the severity of incidents.

Priority Levels

HIGH Priority

Examples:

- Emergency
- Fire
- Blood
- Attack

MEDIUM Priority

Examples:

- Suspicious activity
- Threats
- Unknown individuals

LOW Priority

Examples:

- Minor incidents
- Small accidents

How It Works

The system analyzes:

- Incident category
- Keywords from report description

Then automatically assigns:

- HIGH
- MEDIUM
- LOW

10. Reports Module

The Reports module displays all submitted reports.

Information Displayed

- Incident image
- Report title
- Reporter name
- Category
- Location
- Description
- Status
- Priority level
- Date and time submitted

Report Status

- Pending
- Under Review
- Resolved

11. Notifications Module

The Notifications module informs users about system activities.

Notification Examples

- Report submitted successfully
- Report status updated
- Incident resolved

Notification Badge

A red notification badge appears when there are unread notifications.

Notification Behavior

- Notifications automatically become “read” after viewing.
- Notification count automatically updates.

12. Profile Module

The Profile module displays user information.

Information Included

- Full Name

- Email Address
- Role
- Date Registered
- Contact information
- Student ID

13. Dark Mode and Light Mode

The system supports both dark and light themes.

Features

- User can manually switch themes.
- Theme preference is automatically saved.
- Colors automatically adapt depending on selected mode.

Theme Switching

1. Open Homepage.
2. Toggle the Dark Mode switch.
3. Interface automatically changes.

14. Admin Dashboard

The Admin Dashboard allows administrators to manage the system.

Dashboard Information

- Total Reports
- Pending Reports
- Resolved Reports
- Total Users

Admin Modules

- Manage Reports
- Manage Users
- View Logs
- Logout

15. Manage Reports Module

Admins can monitor all submitted reports.

Admin Features

- View report details
- Open report information
- Monitor priorities
- Update report status

Report Information

- Reporter Name
- Category
- Location
- Description
- Priority
- Uploaded image
- Timestamp

16. Report Status Management

Admins can update incident status.

Available Statuses

- Pending
- Under Review
- Resolved

Process

1. Open report details.
2. Select a status.
3. Save changes.

Automatic Actions

- User receives notification.
- User logs are updated.
- Report status changes in real time.

17. Manage Users Module

Admins can manage registered users.

Features

- View all users
- View user information
- Monitor user activity

Information Displayed

- Full Name
- Email
- Role
- Status
- Date Registered

18. User Logs Module

The User Logs module records system activity.

Logged Activities

- Login
- Logout
- Report submission
- Report status updates

Purpose

Logs help administrators:

- Monitor activity
- Detect misuse
- Improve security

19. Logout

Steps

1. Tap “Logout”.

2. The system signs out the current user.
3. User is redirected to Login screen.

Automatic Action

A logout activity log is automatically created.